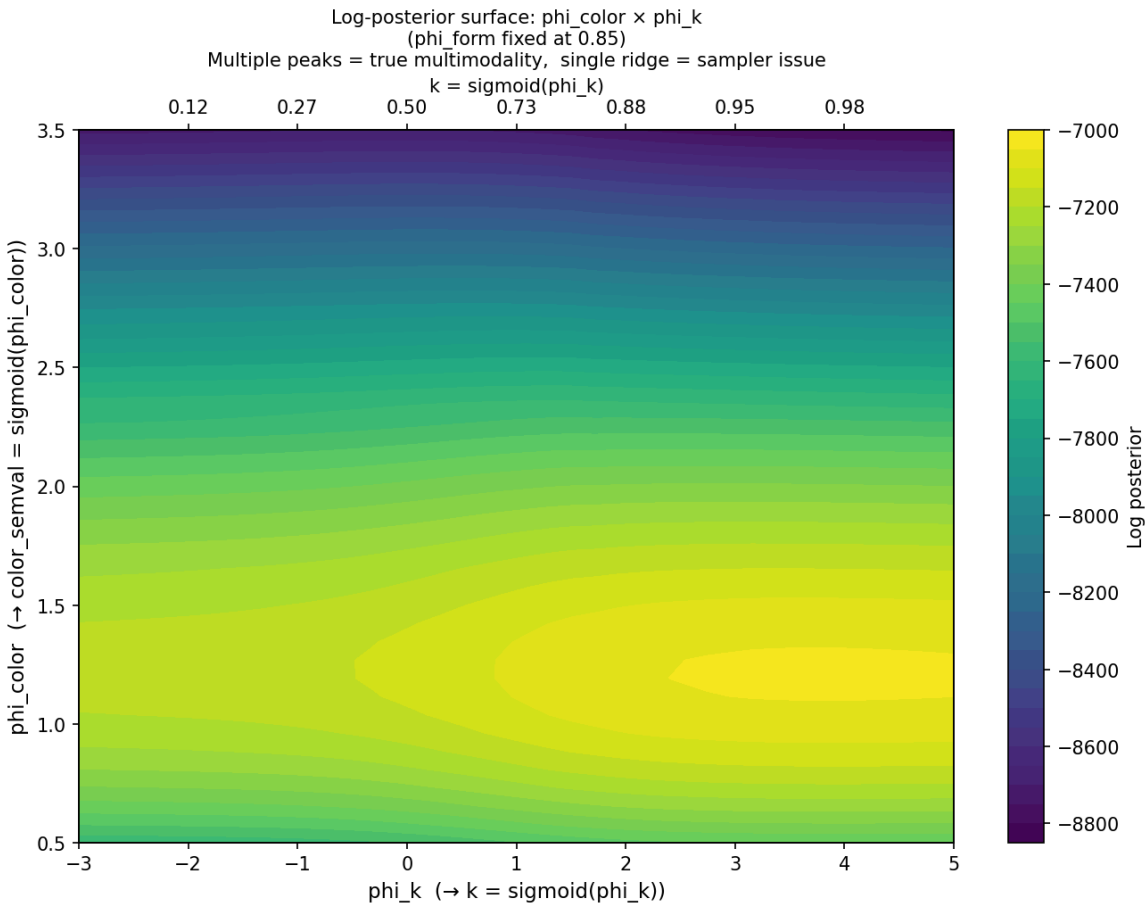
[RSAop] Progress report on modeling with production data

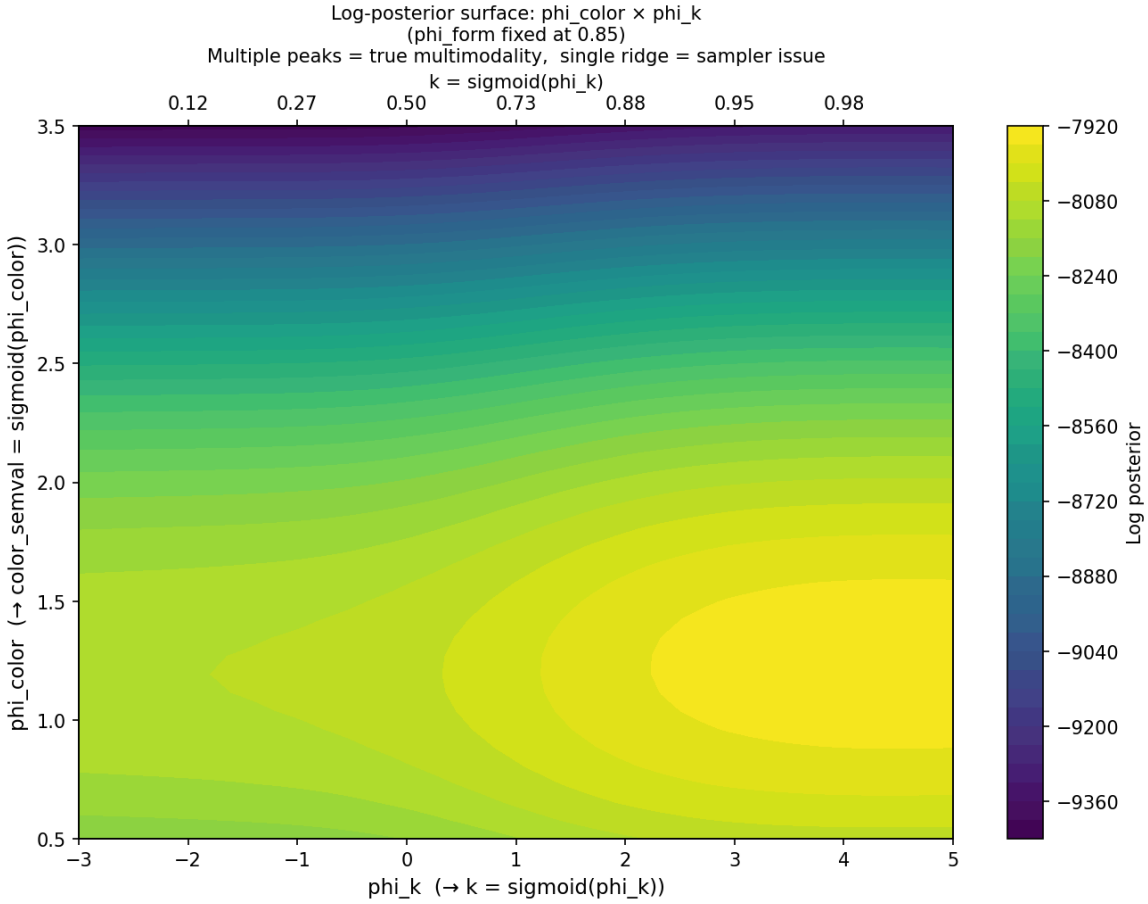
Date: 24-02-2026

Profile log-posterior surface analysis: color and size

Incremental speaker



Global speaker



The profile posterior revealed that **k** (the size threshold position) is not identifiable from the data. This is consistent with two non-exclusive explanations:

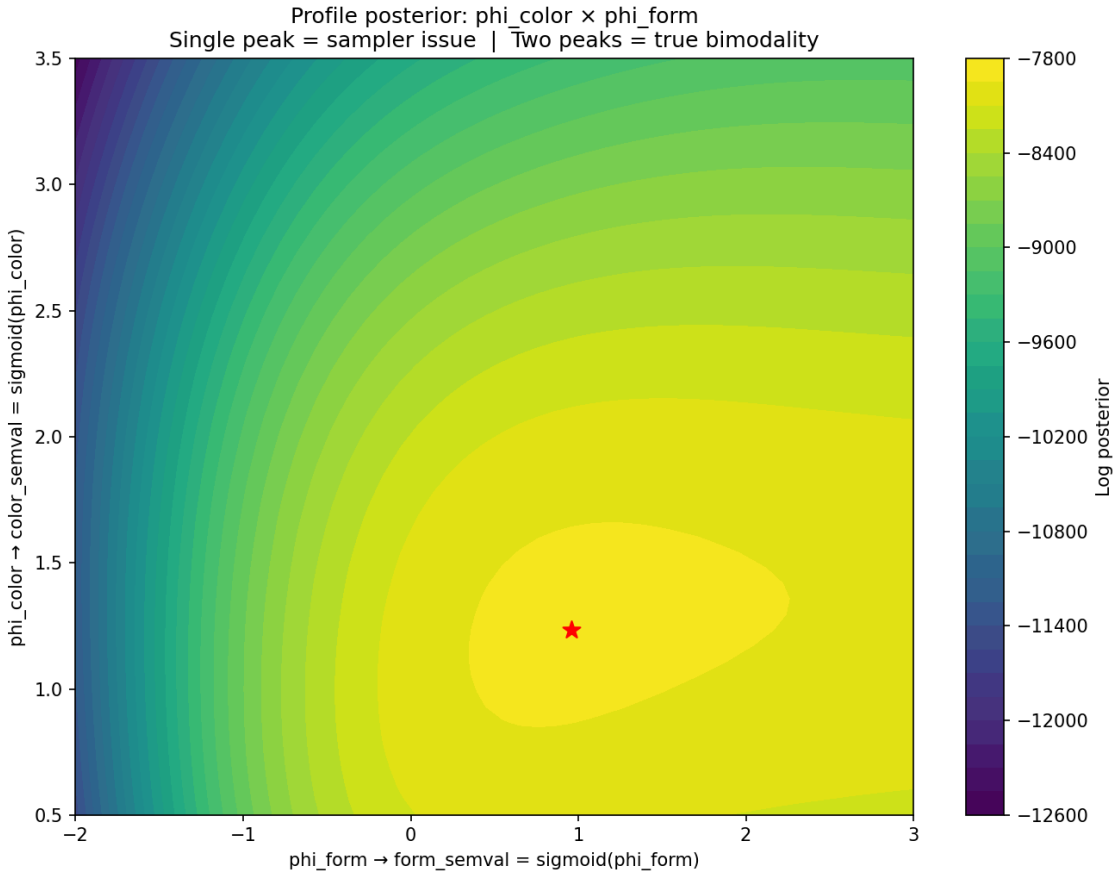
- **(1) Size was not contrastive across trials**, providing no information to locate the threshold.
- **(2) Color and form were sufficiently informative** that size did not contribute additional discriminative signal, making k redundant.

In both cases, the size semantic formula is correctly specified — the threshold $\theta_k = x_{min} + k(x_{max} - x_{min})$ is theoretically sound. The data simply cannot constrain where within the size range the threshold falls.

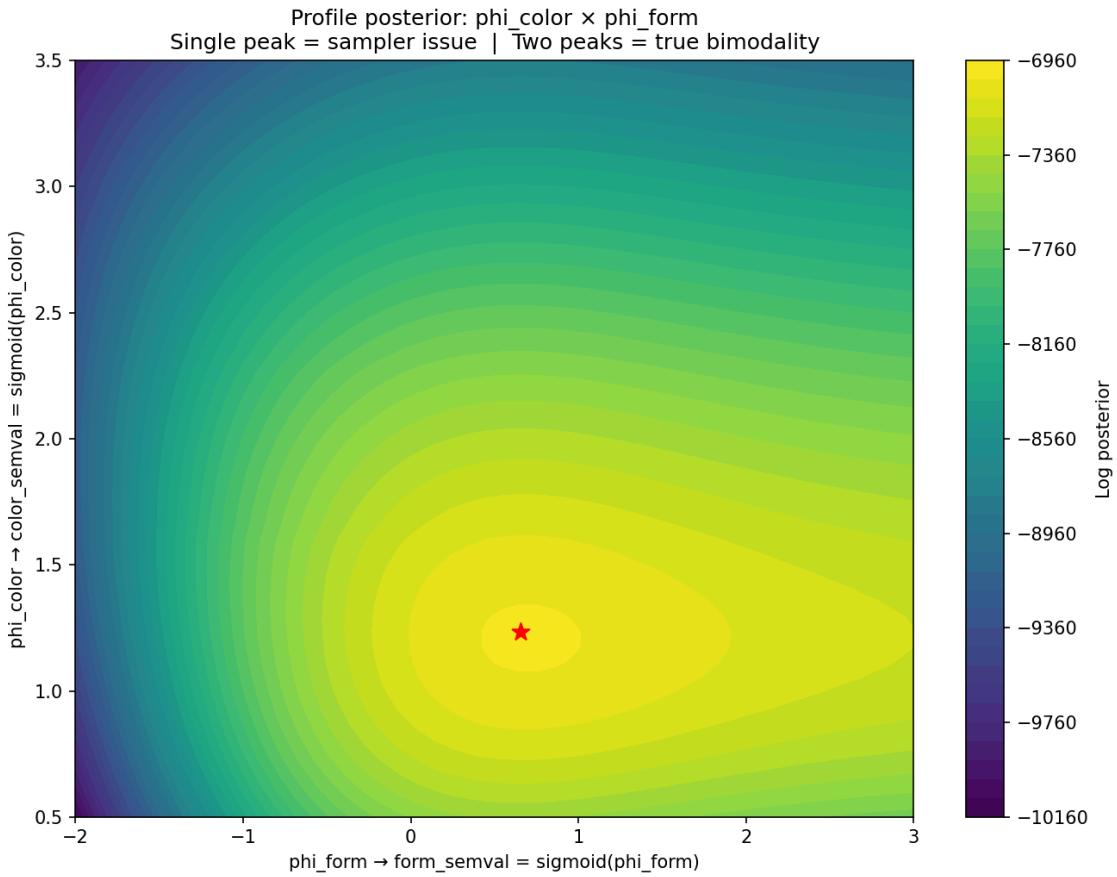
We therefore fixed k = [best_k from grid search] and treated it as a structural constant rather than a free parameter, reducing the model to 2 identifiable parameters (phi_color, phi_form). **Or we adopt another threshold computation method while still keeping it context dependent.**

Profile likelihood analysis: color and form

Global speaker



Incremental speaker



Clearly unimodal. However when we infer them jointly, we get bi-modal posterior, **suggesting issues on the sampler side**. → **which makes the inference not efficient, chain get stuck, posterior results not reliable**

What do we really want here?

Two fundamentally different goals:

GOAL A: Parameter Estimation

"What does the data tell us about color_semval?"

- Infer ALL free parameters
- Want full posterior
- Uncertainty matters

GOAL B: Model Comparison

"Which model better explains human referential behavior?"

- Infer MINIMUM parameters

- Want comparable fits
- Fairness across models matters

I think the goal for our purpose is B, which suggests:

→ **Fix shared meaningful theoretical parameters, infer only parameters controlling noise**

Current parameter settings

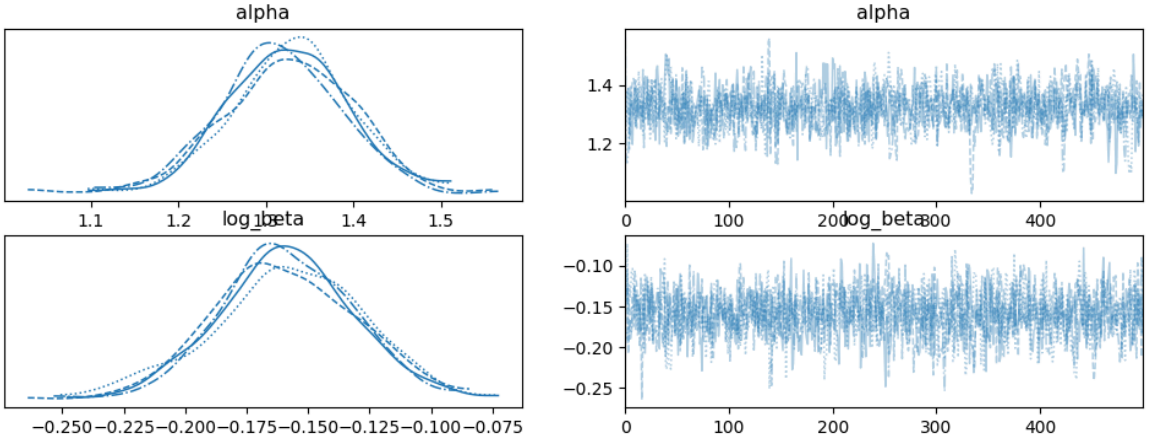
Utility is defined as trade-off between informativity and cost, each governed by a scaling parameter α, β. Costs are elicited for each utterances by a GPT-2 model for German. We only infer α, β.

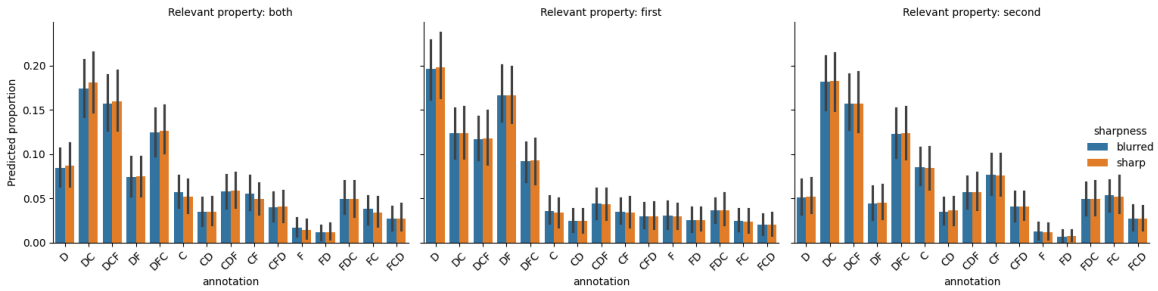
$$U(s^*, u) = \alpha \, \log L(s^* \mid u) \; - \; \beta \, \log P_{\mathrm{LM}}(u),$$

Parameter	Status	Value / Prior	Rationale
alpha	Inferred	halfnormal(5), prior mean around 3 based on the literature	this controls the general noise
beta	Inferred	log_normal(1,0.5), prior mean = 1 based on the previous fit	this controls the LM based utterance cost
k	Fixed	0.5	Size threshold
wf	Fixed	1.0	Weight factor
color_semval	Fixed	0.8 (MAP baseline)	Shared across models
form_semval	Fixed	0.7 (MAP baseline)	Shared across models

Posterior analysis

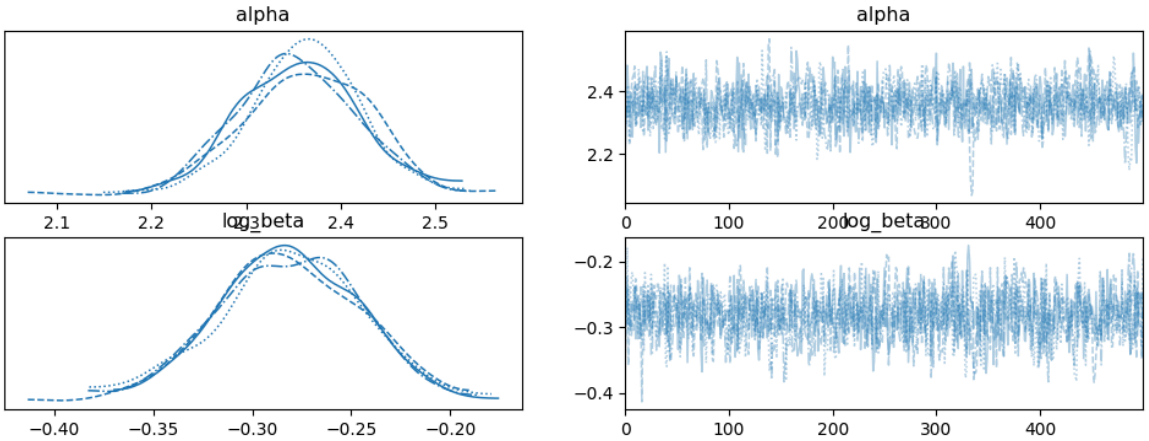
Global speaker



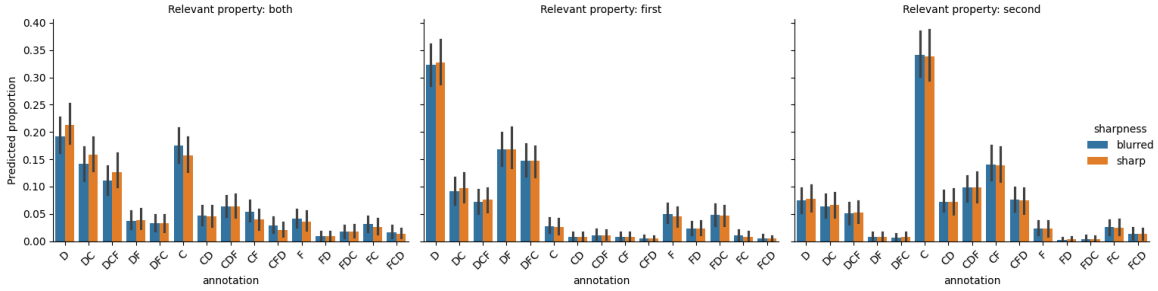


Incremental speaker

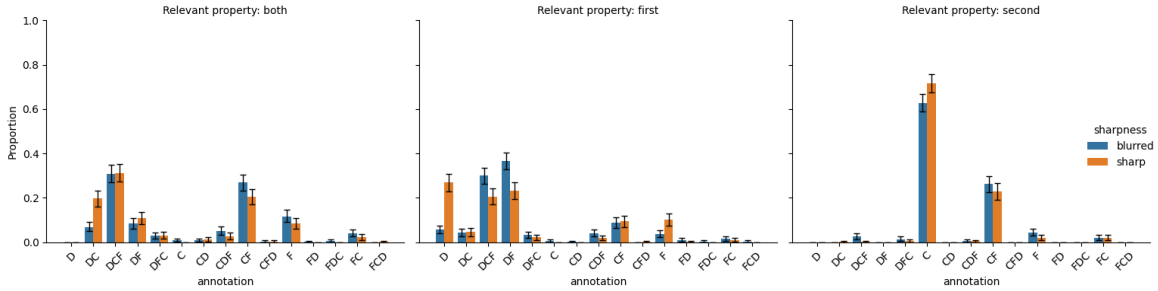
Trace plot



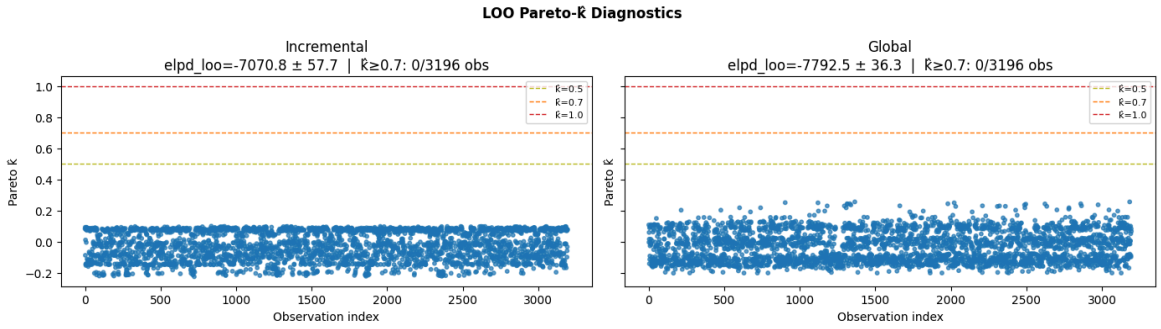
Posterior predictions



Empirical data (for reference)



Model comparison (incremental speaker clearly wins)

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=================================================================
=
MODEL COMPARISON SUMMARY
=================================================================
=
              rank     elpd_loo     p_loo   elpd_diff    wei
ght         se        dse  warning scale
incremental     0 -7070.820572  1.812666    0.000000  0.846
128  57.742861   0.000000    False   log
```

```
global          1 -7792.471197  1.290781  721.650625  0.153
872  36.322019  47.674179    False   log
```

Rationale for a simplified design and more data (or not to do)

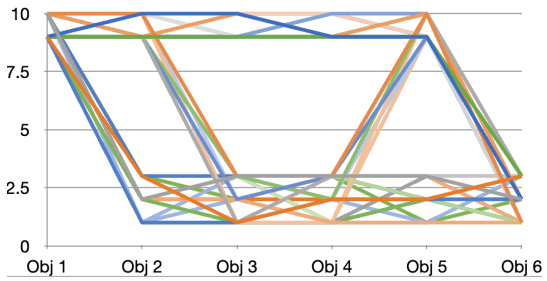
Currently the design:

Three factors: RELEVANCE X COMBINATION X SHARPNESS

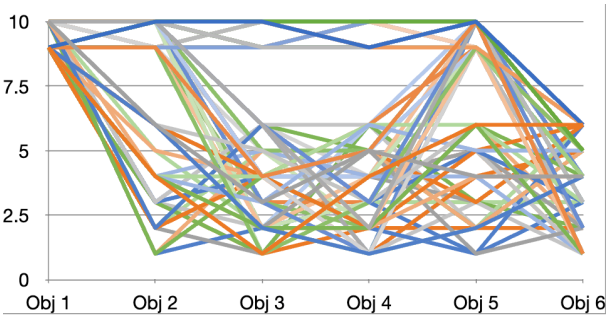
RELEVANCE: first, second, both (corresponds to the coding in COMBINATION)

COMBINATION: color & form, dimension & color, dimension & form

SHARPNESS: sharp, blurred



sharp



blurred

Problems:

1. **We almost never analysed the third property, form, separately, but aggregated it with color into a new dimension & X category** (X is unspecified for color and form), because:
   a. for theoretical reasons: they are similar on the subjectivity scale; we assume the same coin flip, noisy semantics for them
   b. for empirical reasons: when color and form stand in direct comparison, color is preferred early. However, when interacting with size, color and form show similar patterns.
   c. for practical reasons: it is easier to present and to communicate
2. But we made some variations on the third property. While RELEVANCE X COMBINATION controls the manipulation of two properties, the third property (e.g. form in a dimension & color combination) is randomly

assigned following a geometric distribution, which on average makes 3-4 objects be in the same form. But in rare cases (about 10%), it will also create a 5 vs. 1 scenario, which affects the informativity-based reasoning in an unaccounted manner.

→ **There is already a replication without randomly altering third property done.**

3. The model only accounts for the dimension & X category so far, and hence is always fitted to a subset of the data, which marginalizes the factor COMBINATION out. The original thought was to treat color & form as a baseline to show the effect of subjectivity (more subjective adjectives are preferred early). This hypothesis is more like a replication and less central in the current high-level framing (draft) below.
4. 4. The manipulation of size distribution is less sound for not making the differences salient enough. The current manipulation is a little too subtle.
5. The sample size might not be large enough (N = 120 for each experiment respectively)

Suggestions:

Re-run the experiment with only two factors: RELEVANCE, which is framed now as **in-context informativeness** and SHARPNESS, which is framed now as **across context** (or inherent) informativeness. We include only size and color adjectives, and make SHARPNESS more sound by increasing the gap between sizes.

Why not to do:

- A lot of work has been developed so far in a data-driven manner. It is risky when the data tells a different story — the current findings may not survive a new experimental design, making prior efforts harder to justify.
- Costs (money, time, energy ...) — a new experiment requires non-trivial investment, especially if the expected gain in interpretability is marginal.
- Maybe strategic framing can rescue the current design instead? Acknowledging the limitations explicitly and contextualizing them within the broader literature may be sufficient.

High-level framing or summary draft in the form of an abstract

Intro: Speakers of diverse languages exhibit robust ordering preferences of adjectives (e.g., preferring “the big brown bag” over “the brown big bag”).

Theory part: A way to approach these regularities is through communicative efficiency: helping interlocutors coordinate quickly and reliably. Yet “efficiency” can be formalized from two perspectives that make different ordering predictions: a hierarchical, comprehension-oriented view in which modification is composed outward from the noun, and a linear, production-oriented view in which speakers benefit from placing highly informative material early in the unfolding word stream.

Motivation and RQ: Prior work has not fully examined the dynamic pressures of informativeness in online referential communication, specifically the tension between hierarchical structure and linear production. In this study, we investigate how communicative efficiency shapes ordering preferences by manipulating both in-context and across-context informativeness.

Experiments: Across two experiments, we observe a consistent interaction effect: adjectives with high informativeness in both dimensions are preferred earlier in the linear sequence. These findings challenge strict subjectivity-based predictions, which rely on hierarchical structure, and instead support a discriminatory strength hypothesis, where speakers prioritize useful information early in the linear stream.

Models and BDA: To resolve this theoretical conflict, we developed a computational model within the Rational Speech Act framework that contrasts incremental production against global, hierarchical planning. Model comparisons demonstrate that an incremental speaker model best captures human ordering preferences in context.

Simulations: Furthermore, simulations reveal that while incremental pressures drive context-specific variation, both incremental and non-incremental mechanisms can account for the emergence of conventionalized orders over time.

Conclusion: These results offer a unified account of adjective ordering, suggesting that communicative efficiency operates primarily through linear, incremental processing pressures rather than static semantic hierarchies alone.